

GE VERNOVA

Gas Power | Field Services

STANDARD OPERATING PROCEDURE

Borescope Inspection of Heavy-Duty Industrial Gas Turbines (F-Class)

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Revision History

Rev.	Date	Description of Change	Author
A	Mar 14, 2023	Initial release for F-class fleet standardization.	Field Services Engineering
B	Jul 02, 2024	Added factored fired hours (FFH) interval guidance; revised defect disposition table to align with current OEM technical information letters.	Field Services Engineering
C	Aug 09, 2026	Incorporated confined-space entry cross-reference, updated imaging documentation requirements, and clarified acceptance criteria for transition piece aft frame seals.	Field Services Engineering

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1. Purpose

This procedure defines the standard method for performing a borescope inspection of GE Vernova F-Class heavy-duty industrial gas turbines. It establishes the sequence, safety controls, acceptance criteria, and documentation requirements necessary to assess the internal condition of compressor, combustion, and turbine section components without full unit disassembly.

Borescope inspection is a condition-monitoring activity, not a substitute for scheduled combustion or hot gas path (HGP) outages. Its purpose is early detection of service-induced wear or damage so that maintenance planning, interval optimization, and outage scope decisions can be made on a condition basis, within OEM-approved limits.

2. Scope

This SOP applies to borescope inspections performed on GE Vernova 7F- and 9F-platform heavy-duty industrial gas turbines at sites operating under a GE Vernova service agreement or long-term parts and services agreement (LTPSA). It covers compressor, combustion, and turbine section (1st through 3rd stage) internal visual inspection using video borescope equipment.

This procedure does not cover generator internal inspections, off-line water wash, aircraft-derivative (LM-series) turbines, or teardown-level combustion and HGP inspections requiring component removal — these are addressed in separate governing procedures.

3. References and Applicable Standards

- GE Vernova Gas Turbine Maintenance and Safety Manual (unit-specific, per frame and controls platform)
- Applicable GE Vernova Technical Information Letters (TILs) for the affected fleet/frame
- Site-specific Lockout/Tagout (LOTO) procedure
- Site-specific Confined Space Entry procedure
- OSHA 29 CFR 1910.146 — Permit-Required Confined Spaces
- OSHA 29 CFR 1910.147 — The Control of Hazardous Energy (Lockout/Tagout)
- ASNT SNT-TC-1A — Personnel Qualification and Certification in Nondestructive Testing (borescope operator qualification basis)
- Site Environmental, Health, and Safety (EHS) PPE matrix

4. Definitions and Acronyms

Term	Definition
Borescope	Optical or video inspection instrument inserted through dedicated ports to visually examine internal turbine components without disassembly.
FFH	Factored Fired Hours — an equivalent-hours metric that weights operating hours by duty-cycle severity (starts, trips, fuel type) to normalize maintenance intervals.
HGP	Hot Gas Path — the combustion and turbine section components exposed to the primary gas stream.

LOTO	Lockout/Tagout — hazardous energy control procedure applied before entry or internal work.
TIL	Technical Information Letter — OEM-issued fleet guidance addressing a specific technical condition, inspection requirement, or design update.
Disposition	The accept/reject classification assigned to an observed indication based on acceptance criteria.

5. Roles and Responsibilities

Role	Responsibilities
Site Lead Technician	Owns execution of the inspection; verifies isolation and permits prior to work start; confirms borescope calibration status; signs off on completed inspection report.
Certified Borescope Operator	Performs the physical inspection; operates borescope hardware; captures and annotates imagery; records findings against acceptance criteria in real time.
Mechanical Maintenance Engineer	Reviews recorded findings; classifies indications per the defect disposition table; determines whether findings require corrective action, re-inspection interval adjustment, or engineering evaluation.
Plant Operations / Outage Coordinator	Coordinates unit availability, cooldown timing, and access windows; issues and closes work permits; manages competing outage-schedule priorities.
EHS Representative	Verifies lockout/tagout, confined space, and hot work permits are in place before entry; may witness entry for confined-space classified inspections.
Reliability Engineer (fleet)	Trends findings against fleet history and Technical Information Letters (TILs); recommends interval adjustments based on Factored Fired Hours (FFH) consumption.

6. Safety Precautions

All personnel performing or supporting this procedure must comply with site EHS requirements in addition to the precautions below. This section is not a substitute for site-specific permits and job hazard analysis.

WARNING

Do not open borescope ports or begin inspection until the unit has been confirmed de-energized, isolated per LOTO, and cooled to a safe handling temperature per the site cooldown log. Residual case temperatures can cause severe burns.

CAUTION

Turbine casings and enclosures may qualify as permit-required confined spaces depending on site configuration. Confirm classification with the EHS representative before entry and follow the site confined space entry procedure, including atmospheric monitoring and attendant requirements.

CAUTION

Never force a borescope insertion tube against resistance. Forced insertion can damage the instrument and, more critically, internal turbine components. Withdraw and reassess the approach angle.

NOTE

Rotor barring must remain under positive control at all times during inspection. Confirm with the site lead technician before manually rotating the rotor between borescope stages.

Additional precautions:

- Wear all PPE specified on the active work permit, including hearing protection where turning gear or adjacent equipment is operating.
- Verify gas detection meter calibration and perform a bump test before any confined-space or enclosure entry.
- Maintain communication with the site lead technician throughout the inspection, particularly during rotor barring and port access.
- Stage borescope insertion tubes and cabling to avoid trip hazards and to prevent contact with hot surfaces.

7. Required Tools and Equipment

Item	Specification / Notes
Video borescope system	Articulating, 6.0–6.2 mm insertion tube (compressor/turbine ports); minimum 4-way tip articulation; measurement (stereo or shadow) capability for crack and clearance sizing.
Rigid/semi-rigid mirror scope	For select combustor and first-stage nozzle ports where line-of-sight access permits.
Calibrated measurement tips	Current calibration certificate on file; verify calibration due date before dispatch.
Port plug removal tool kit	Site- and frame-specific socket/wrench set for borescope port plugs and covers.
Dry, oil-free compressed air source	For port cleaning and minor debris clearing only — not for personnel cooling in confined spaces.
Turning gear / rotor barring equipment	To index rotor to required clock positions between stages.
Digital imaging workstation	For on-site image review, annotation, and report generation; synced to fleet reliability database.
PPE	Per site EHS matrix: hearing protection, safety glasses, FR clothing, hard hat; additional PPE per confined-space and hot work permits.
Gas detection meter	Calibrated multi-gas monitor (O ₂ , CO, LEL, H ₂ S) where confined-space or enclosure entry is required.
Approved work permits	LOTO permit, confined space entry permit (if applicable), hot work permit (if applicable), site access/outage work order.

8. Pre-Inspection Preparation

1. Confirm the unit is out of service, isolated per LOTO, and has reached safe handling temperature per the site cooldown log.
2. Verify all required work permits (LOTO, confined space, hot work if applicable) are issued and posted at the work location.
3. Retrieve the unit's borescope inspection history and most recent findings for trend comparison.
4. Inspect borescope hardware for physical damage; confirm power-on self-test passes and measurement tip calibration is current.
5. Confirm gas detection meter calibration and complete bump test if enclosure or confined-space entry is required.
6. Coordinate with the site lead technician to confirm turning gear/rotor barring readiness and communication protocol for rotor positioning.
7. Review the applicable frame-specific port map and rotor clock-position sequence for the unit being inspected.
8. Stage the digital imaging workstation and confirm inspection report template is loaded with correct unit identification.

9. Inspection Procedure

9.1 Compressor Section

1. Remove inlet screen or access panel per unit configuration; locate designated compressor borescope ports per the frame-specific port map.
2. Remove port plugs in sequence, capping open ports not currently in use to prevent debris ingress.
3. Insert the borescope and inspect each compressor stage for leading- and trailing-edge damage, foreign object damage (FOD), erosion, corrosion pitting, and coating condition.
4. Bar the rotor in accordance with the site turning gear procedure to index each blade row past the port; capture representative images at each indexed position and at any location showing an indication.
5. Progress through all compressor stages per the port map, working from the inlet toward the discharge.

9.2 Combustion Section

1. Access combustion liner and transition piece ports per the frame-specific port map; remove plugs in sequence.
2. Inspect each combustion liner for cracking, distortion, coating condition, and cooling hole blockage.
3. Inspect each transition piece for cracking, distortion, and aft frame seal wear; assess seal gap condition against acceptance criteria.
4. Inspect fuel nozzle tips visible from the accessible port for coking, erosion, or physical damage.
5. Repeat for each can/combustor in the unit's configuration; capture images referenced to can number and port position.

9.3 Turbine Section (Hot Gas Path)

1. Access 1st-stage nozzle and bucket ports per the frame-specific port map.
2. Inspect the 1st-stage nozzle for cracking, coating spallation, thermal barrier coating (TBC) loss, and cooling hole condition.

3. Bar the rotor to index each 1st-stage bucket past the port; inspect leading and trailing edges, platform, tip, and coating condition at each position.
4. Repeat the applicable inspection sequence for 2nd- and 3rd-stage nozzles and buckets per the frame's available port access.
5. Pay particular attention to areas flagged in prior inspection history or identified in applicable Technical Information Letters for the fleet.

9.4 Photographic Documentation

1. Capture at minimum one representative image per stage/component per rotor position inspected, and additional images for any indication.
2. Annotate each image at the time of capture with component, stage, rotor clock position, and port reference.
3. Log every recorded indication in the photographic log (Appendix B), cross-referenced to the acceptance criteria table.
4. Back up all images and the draft inspection report to the fleet reliability database before demobilizing from site.

10. Acceptance Criteria and Defect Disposition

Findings are classified as Accept or Reject based on the criteria below. This table summarizes common indications; it does not replace unit- and component-specific limits published in the applicable OEM technical manual and current Technical Information Letters, which govern in case of any conflict.

Component	Indication Type	Disposition Category	Required Action
Compressor blades (all stages)	Minor leading-edge nicks/dents within OEM dimensional limits	Accept	Log; re-inspect at next scheduled interval
Compressor blades (all stages)	Nicks/dents exceeding blend-limit dimensions; foreign object damage	Reject	Ground-blend evaluation by engineering; possible interval reduction
Combustion liner / transition piece	Localized discoloration, minor coating wear	Accept	Log; photograph for trend comparison
Combustion liner / transition piece	Through-wall indication, cracking beyond OEM limit, missing hardware	Reject	Immediate notification to reliability engineering; unit derate or outage evaluation
Transition piece aft frame seals	Seal wear within allowable gap tolerance	Accept	Log; monitor trend
Transition piece aft frame seals	Seal wear exceeding allowable gap tolerance	Reject	Schedule seal replacement at next outage
1st-stage turbine nozzle	Minor trailing-edge erosion, coating loss within limits	Accept	Log; re-inspect at next scheduled interval
1st-stage turbine nozzle	Cracking, significant coating spallation, TBC loss exposing	Reject	Engineering evaluation; life assessment required

	substrate		
1st/2nd-stage turbine buckets	Minor tip rub, coating discoloration	Accept	Log; photograph for trend comparison
1st/2nd-stage turbine buckets	Crack indication, missing tip material, platform distress	Reject	Immediate engineering notification; unit run restrictions per TIL guidance

NOTE

Any indication that cannot be confidently classified using this table must be escalated to the Mechanical Maintenance Engineer for disposition before the unit is returned to service.

11. Post-Inspection Activities

1. Reinstall all borescope port plugs and torque to the specification in the unit's maintenance manual.
2. Confirm no tools, plugs, or foreign material remain inside the casing or on adjacent surfaces before closing access panels.
3. Disengage turning gear/rotor barring equipment per the site procedure and confirm the unit is in an operations-ready mechanical state.
4. Complete the inspection report, including all findings, dispositions, and recommended follow-up actions; route to the Mechanical Maintenance Engineer for review.
5. Update the unit's borescope inspection history and fleet reliability database with current findings and recommended next inspection interval.
6. Close out all work permits with the site lead technician and Operations.

12. Summary of Notes, Cautions, and Warnings

WARNING

Confirm LOTO isolation and safe casing temperature before any port access. Do not bypass cooldown requirements to meet an outage schedule.

CAUTION

Treat all turbine enclosures as potential confined spaces until classified otherwise by the site EHS representative.

NOTE

Escalate any indication outside the scope of the acceptance criteria table to engineering before returning the unit to service.

Appendix A — Borescope Inspection Checklist

To be completed in sequence during each inspection and retained with the inspection report.

✓	Checklist Item	Initials / Time
☐	Work permits (LOTO, confined space, hot work) issued and posted	
☐	Unit confirmed at safe handling temperature per cooldown log	
☐	Gas detection meter bump-tested and calibration current	
☐	Borescope system powered on; self-test passed; measurement tip calibration verified	
☐	Turning gear engaged and rotor barring function confirmed	
☐	Compressor inlet/bellmouth borescope ports located and plugs removed	
☐	Compressor stages inspected per rotor clock-position sequence; images captured	
☐	Combustion liners and transition pieces inspected at each can; images captured	
☐	1st-stage turbine nozzle inspected; images captured	
☐	1st-stage turbine buckets inspected at each rotor position; images captured	
☐	2nd/3rd-stage turbine components inspected per applicable frame scope	
☐	All indications logged against acceptance criteria table	
☐	Photographic log cross-referenced to port and rotor position	
☐	Port plugs reinstalled and torqued to specification	
☐	Turning gear disengaged per site procedure; unit returned to operations-ready state	
☐	Inspection report drafted and findings communicated to reliability engineering	
☐	Work permits closed out	

Appendix B — Photographic Findings Log

Use in conjunction with Section 9.4. Attach corresponding image files referenced by Image # to the digital inspection report.

Image #	Component	Port / Location	Finding Description	Disposition
1				
2				
3				
4				
5				
6				
7				
8				